

Akansha Rautela

Research Associate,
Department of Electrical Engineering
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Research Interests

- Optimal Control; Multi-Agent Systems; Distributed and Networked Control; Constrained Control; Energy/Fuel and Time-Optimal Consensus; State Estimation.

Education

Ph.D. Electrical Engineering

Sept 2020 - July 2026

Indian Institute of Technology Delhi

New Delhi

- **Thesis:** Computation of Optimal Consensus in Multi-Agent Systems Using Gröbner Basis
- **Advisors:** Prof. Deepak U. Patil and Prof. Indra Narayan Kar

M.Tech. Instrumentation

May 2019

National Institute of Technology Kurukshetra

Haryana

- **Thesis:** Line of Sight (LOS) Stabilization using Disturbance Observer (DOB)

B.Tech. Applied Electronics and Instrumentation Engineering

May 2015

College of Engineering Roorkee

Roorkee

Academic and Research Experience

Research Associate

April 2026 - Present

Indian Institute of Technology Delhi

New Delhi

- **PIs:** Prof. Indra Narayan Kar and Prof. Arpan Chattopadhyay
- **Project:** Improved Bearings-only Target Motion Analysis (BO-TMA) Using AI Tools (IFCPAR funded)
- Analyzing target motion dynamics utilizing Set-Membership BO-TMA approaches to bound estimation errors under uncertainty.

Teaching Assistant

Sept 2020 - Dec 2025

Indian Institute of Technology Delhi

New Delhi

- **Courses:** (ELL333) Multivariable Control; (ELL705) Stochastic Filtering and Identification; (ELL225 & ELP225) Control Engineering (Theory & Lab); (ELP101) Introduction to Electrical Engineering Lab.
- Graded assignments and guided students in laboratory experiments.

Research Intern (Servo Lab)

Jan 2019 - Jul 2019

Defence Research and Development Organisation (IRDE)

Dehradun

- Designed and simulated a Disturbance Observer (DOB) in MATLAB/Simulink for Line of Sight (LOS) stabilization under external disturbances.

Publications and Research Work

Journals

- Akansha Rautela, Deepak Patil, Ameer Mulla, and I. N. Kar, “Minimum Time Consensus of Multi-agent System under Fuel Constraints.” *IEEE Transactions on Automatic Control*, 2026. (3rd revision).
- Akansha Rautela, Deepak Patil, Ameer Mulla, and I. N. Kar, “Minimum Fuel Budget Allocations for Consensus in Multi-agent Systems.” (In preparation).
- Akansha Rautela, Deepak Patil, Ameer Mulla, and I. N. Kar, “Minimum time consensus for damped second order agents using Gröbner basis.” 2026. (In preparation).
- Akansha Rautela, Deepak Patil, Ameer Mulla, and I. N. Kar, “Minimum Time Consensus of Multi-agent System under Energy Constraints using Attainable Sets.” 2026. (In preparation).

Conferences and Extended Abstracts

- Akansha Rautela, Deepak Patil, Ameer Mulla, and I. N. Kar, “Minimum time consensus for damped second order agents using Gröbner basis.” 65th *IEEE Conference on Decision and Control (CDC)*, 2026. (Accepted).
- Akansha Rautela, Deepak Patil, Ameer Mulla, and I. N. Kar, “Minimum Time Consensus of Multi-agent System under Energy Constraints.” presented at 23rd *IFAC World Congress*, Busan, Republic of Korea, August 23-28, 2026.
- Akansha Rautela, Deepak Patil, Ameer Mulla, and I. N. Kar, “Minimum Fuel Consensus for Multi-agent System.” *Joint IFAC Conference SSSC-TDS-COSY*, Gif-sur-Yvette, France, July 2025. DOI: <https://doi.org/10.1016/j.ifacol.2025.09.518>.
- Akansha Rautela, Deepak Patil, Ameer Mulla, and I. N. Kar, “Minimum Time Consensus for a Class of Second Order LTI Systems Under Fuel Constraints.” *Joint IFAC Conference SSSC-TDS-COSY*, France, July 2025.
- Akansha Rautela, Deepak Patil, Ameer Mulla, and I. N. Kar, “Minimum Time Consensus of Multi-agent System under Fuel Constraints.” 26th *International Symposium on Mathematical Theory of Networks and Systems (MTNS)*, University of Cambridge, UK, Aug 2024.
- Akansha Rautela, and R. P. Chauhan, “Line-of-sight (LOS) stabilization using a disturbance observer (DOB).” *International Conference on Sustainable Communication Networks and Application*, Springer, 2019. DOI: https://doi.org/10.1007/978-3-030-34515-0_81.

Talks and Presentations

- “Computing Minimum Time Consensus of Multi-agent System under Energy Constraints using Gröbner Basis” presented at 23rd IFAC World Congress in Busan, Republic of Korea (August 23-28, 2026).
- “Computation of optimal consensus in multi-agent system using Gröbner Basis” at the PhD Forum, 23rd IFAC World Congress, Busan, Republic of Korea (August 23 - 28, 2026).
- “Minimum Fuel Consensus for Multi-agent System”, presented at the *Joint IFAC Conference SSSC-TDS-COSY*, CentraleSupélec, Gif-sur-Yvette, France, July 2025.
- “Minimum Time Consensus of Multi-agent System under Fuel Constraints”, presented at the 26th *International Symposium on MTNS*, University of Cambridge, UK, August 2024.

Technical Skills

- **Languages & Tools:** Scilab, MATLAB, Python, SageMath, Simulink, ROS2.
- **Documentation:** L^AT_EX, Overleaf, MS Office.

Workshops and Courses

- GIAN Course on Data Based Systems and Control, IIT Mandi (April 2025).
- Workshop on Systems, Learning and Control, Control Group, EE Dept, IIT Delhi (Jan 2025).
- Short Term Course on Data Analytics, NIT Kurukshetra (June 2019).
- Short Term Course on Estimation Guidance and Control: Application to Missile Systems, NIT Kurukshetra under TEQIP-III (Jan 2018).

Professional Activities & Awards

- **Reviewer:** IEEE Transactions on Automatic Control, IEEE Transactions on Aerospace and Electronic Systems; IEEE CDC 2026; IFAC World Congress 2026; Joint SSSC, TDS, COSY 2025; IEEE Indian Control Conference.
- **Scholarships:** MoE GATE scholarship for M.Tech at NIT Kurukshetra, MoE scholarship for PhD at IIT Delhi.
- **Awards:**
 - 1) Certificate of Distinction in PhD research awarded to the finalists of the PhD Forum at IFAC World Congress 2026.
 - 2) IFAC D&I Travel Support to attend 23rd IFAC World Congress in Busan, Republic of Korea, August 23–28, 2026.
 - 3) Selected for IFAC Wearable Robot Challenge in IFAC World Congress 2026.

Referees

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| <p>1. Prof. Deepak U. Patil
II-402B, Department of Electrical Engineering,
Indian Institute of Technology Delhi, India
Email: deepakpatil@ee.iitd.ac.in
Phone: +91-11-26591094</p> | <p>3. Prof. Subashish Datta
Department of Electrical Engineering,
Indian Institute of Technology Delhi, India
Email: subashish@ee.iitd.ac.in
Phone: +91-11-26591077</p> |
| <p>2. Prof. Indra Narayan Kar
Department of Electrical Engineering,
Indian Institute of Technology Delhi, India
Email: ink@ee.iitd.ac.in
Phone: +91-11-26591093</p> | <p>4. Prof. Ameer K Mulla
1031 Academic Block-I (Engineering) A1 - 323,
Indian Institute of Technology Dharwad, India
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